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ENGINEERING
TETE/TEEE 3102: MICROCONTROLLERS

LAB REPORT
SMART ROOM MONITORING SYSTEM
Level 2 Upgrade: Detect, Evidence, Alert & Power
Monitor

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1 Introduction: Upgrade from Level 1

The initial implementation (Level 1) of the Smart Room Monitoring System focused on basic monitoring, local alerts via a buzzer, and data logging to ThingSpeak.

Level 2 significantly upgrades the system capabilities to encompass:

- **Detection:** Advanced intrusion and safety hazard detection.
- **Evidence Capture:** Image capturing capabilities upon motion detection.
- **Alert Anywhere:** Reliable SMS/Call alerts even when Wi-Fi is down.
- **Power Monitoring:** Comprehensive tracking of power consumption and grid status.

2 New Hardware Additions

To achieve the Level 2 functionalities, the following hardware components have been integrated into the system, bringing the total number of sensors to 7 (DHT22, Ultrasonic, PIR, Camera, Voltage, Current, MQ-2, Reed Switch).

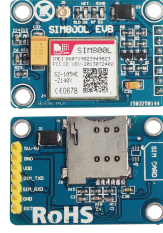
1. **ESP32-CAM (Replacing Arduino Uno/ESP8266):** Utilized for image capture when motion is detected. It has built-in Wi-Fi, efficiently acting as the primary web-connected controller.
2. **SIM800L GSM Module:** Critical for the Ugandan context. It sends SMS or initiates phone calls when Wi-Fi is unavailable or for critical security/safety alerts.
3. **Voltage Sensor (ZMPT101B) & Current Sensor (ACS712 20A):** Added to monitor the UEDCL mains. These detect blackouts, over-voltage scenarios, and calculate power consumption in kWh.
4. **MQ-2 Gas Sensor:** Deployed in the kitchen area for LPG and smoke leak detection.
5. **Magnetic Reed Switch:** Placed on the main door/window for reliable intrusion detection (acting as a robust complement to the PIR sensor).

3 Hardware Component Visuals

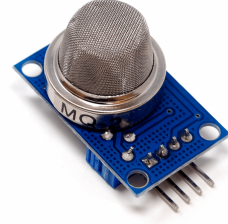
The physical hardware modules integrated into the Level 2 monitoring system are illustrated below:



(a) ESP32-CAM Module



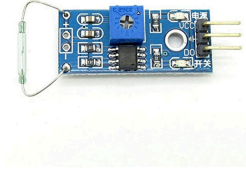
(b) SIM800L GSM Module



(c) MQ-2 Gas Sensor



(d) ZMPT101B AC Voltage Sensor



(e) Magnetic Reed Switch Module

Figure 1: New Hardware Components for Level 2 Upgrade

4 System Requirements & Architecture

The system architecture is divided into four distinct functional layers to ensure robust monitoring and alerting.

4.1 A. Security Layer

- If the **Reed Switch** is OPEN or **PIR** = 1 after 10:00 PM, the ESP32-CAM captures an image.
- The image is saved to the SD Card and uploaded to a Telegram Bot.
- Simultaneously, the GSM module sends an SMS: “*INTRUSION at Door 1 - Time: [HH:MM]*”.

4.2 B. Power Monitoring Layer

- The system continuously reads voltage and current.
- If $Voltage < 180V$, $Voltage > 260V$, or $Voltage = 0$ (blackout): Sends an SMS and a ThingSpeak alert.
- Calculations: $Power(W) = V \times I$. Cumulative Energy (kWh) is calculated and posted to ThingSpeak Fields 5, 6, and 7.

4.3 C. Safety Layer

- If **MQ-2** readings exceed the safety threshold (gas leak detected), the Buzzer turns ON continuously.

- The GSM module initiates a phone call.
- A relay is activated to trigger an exhaust fan or open a window.

4.4 D. Data Layer

- The **ThingSpeak Channel** is upgraded to utilize 8 fields: Temperature, Humidity, Distance, Motion, Voltage, Current, Power, and Gas Level.
- **Blynk** or **Telegram** is implemented for real-time control and system status queries.

5 System Block Diagram

Below is the block diagram representing the data flow and interconnections between the 7 sensors, the core processing unit, and the output interfaces.

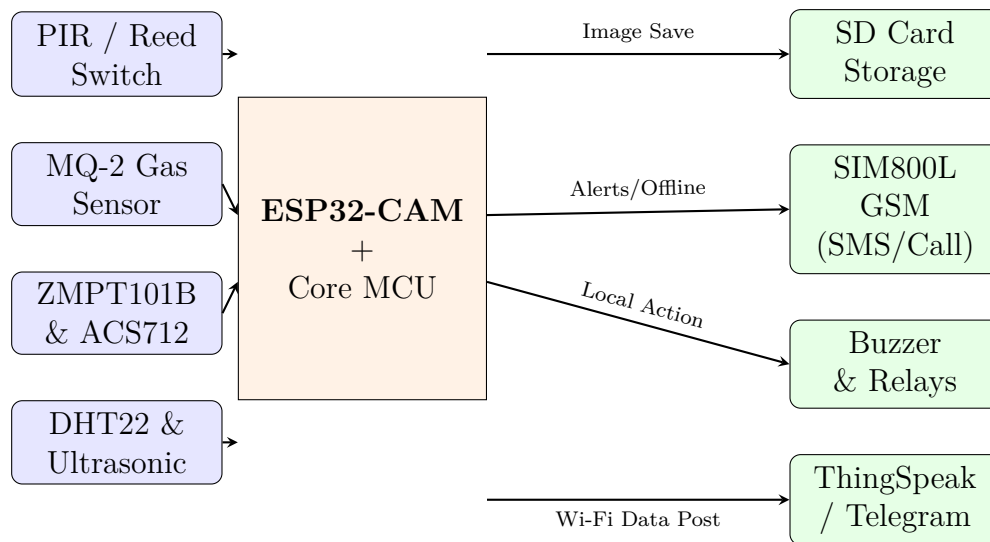


Figure 2: Level 2 Architecture Block Diagram

6 Deliverables

The finalized project submission includes the following deliverables:

1. **Block Diagram & Fritzing Circuit:** Detailed schematic and architectural overview.
2. **Working Prototype:** Physical assembly of all 7 sensors, GSM, and ESP32-CAM.
3. **Live Dashboard Link:** ThingSpeak and Telegram bot access containing at least 48 hours of live logged data.
4. **Costing Analysis:** A Bill of Materials (BOM) cost estimation for a hypothetical deployment of 100 units.